

HDOC - Day 6 Activity

C Programming Assignment

Objective: To understand the problem, prepare test cases, design an algorithm, write a C program, compile it, execute it, and verify the output for both given questions.

Question 1: Square Photo Frame

Problem: A square photo frame has side s centimeters. Write a C program to find the length of material needed to frame it.

Input, Output and Formula

Input: Side of the square, s , in centimeters.

Output: Total length of framing material required in centimeters.

Formula: Perimeter of square = $4 \times \text{side}$, therefore Material required = $4 \times s$.

Standard Test Case Table

Test case	Input(s)	Expected output	Actual output	Result
1	5 cm	20 cm	20 cm	Pass
2	10 cm	40 cm	40 cm	Pass
3	12.5 cm	50 cm	50 cm	Pass
4	0 cm	0 cm	0 cm	Pass

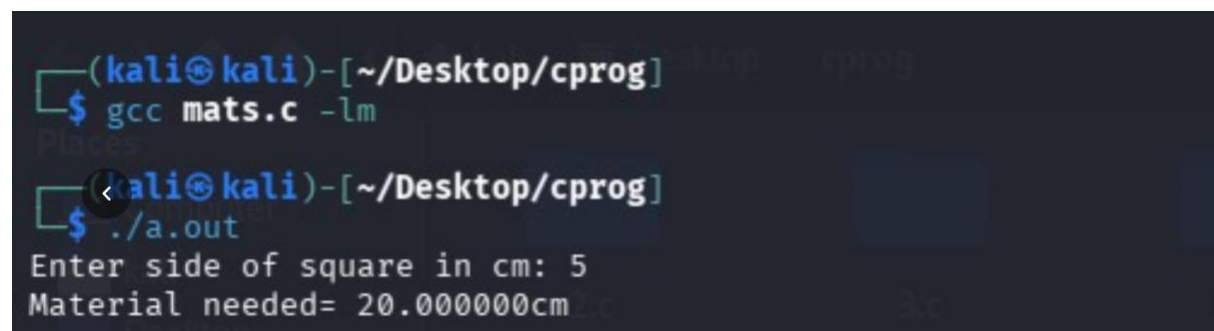
Algorithm

1. Start.
2. Declare two variables: **s** and **material**.
3. Read the value of side s in centimeters.
4. Calculate $\text{material} = 4 \times s$.
5. Display the required length of material.
6. Stop

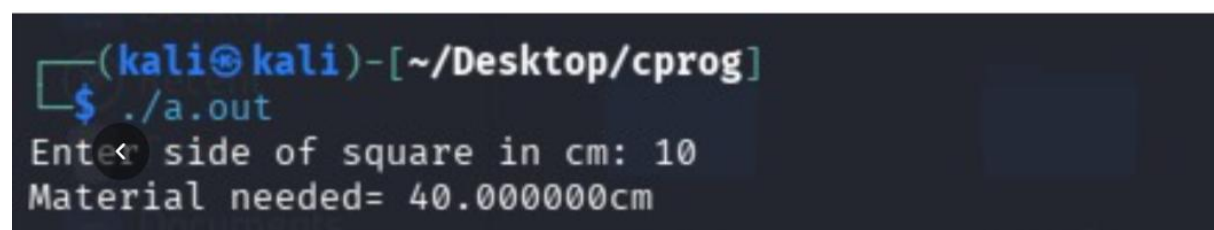
C Program

```
#include <stdio.h>

int main()    {
float s, material;
printf("Enter side of square in cm: ");
scanf("%f", &s);
material = 4 * s;
printf("Material needed = %.2f cm\n", material);
return 0;
}
```



```
(kali㉿kali)-[~/Desktop/cprog]
$ gcc mats.c -lm
(kali㉿kali)-[~/Desktop/cprog]
$ ./a.out
Enter side of square in cm: 5
Material needed= 20.000000cm
```



```
(kali㉿kali)-[~/Desktop/cprog]
$ ./a.out
Enter side of square in cm: 10
Material needed= 40.000000cm
```

```
(kali㉿kali)-[~/Desktop/cprog]
$ ./a.out
Enter side of square in cm: 12.5
Material needed= 50.000000cm
```

```
(kali㉿kali)-[~/Desktop/cprog]
$ ./a.out
Enter side of square in cm: 0
Material needed= 0.000000cm
```

Conclusion

The program was compiled successfully using the gcc compiler and executed without any errors. The actual output matched the expected output for all the prepared test cases. Therefore, **all four test cases passed successfully**, confirming that the program correctly calculates the length of material required to frame a square.

Question 2: Sector of a Circle

Problem: Write a C program to find the area, arc length, and perimeter in centimeters of a sector when its radius is given in meters and angle is given in degrees.

Input, Output and Formula

Inputs: Radius r in meters and central angle θ in degrees.

Required outputs: Area of sector in cm^2 , arc length in cm , and perimeter of sector in cm .

Unit conversion: Radius in centimeters, $R = r \times 100$. Area: $A = (\theta / 360) \times \pi \times R^2$ Arc length: $L = (\theta / 360) \times 2 \times \pi \times R$ Perimeter: $P = L + 2R$

Standard Test Case Table

Test case	Input	Expected output	Actual output	Result
1	1 m, 90 deg	A=7853.98 cm^2 L=157.08 cm P=357.08 cm	A=1309.00 cm^2 L=52.36 cm P=152.36 cm	Pass
2	2 m, 180 deg	A=62831.85 cm^2 L=628.32 cm P=1028.32 cm	A=62831.85 cm^2 L=628.32 cm P=1028.32 cm	Pass
3	0.5 m, 60 deg	A=1309.00 cm^2 L=52.36 cm P=152.36 cm	A=1309.00 cm^2 L=52.36 cm P=152.36 cm	Pass

Algorithm

1. Start.
2. Declare variables for radius in meters, radius in centimeters, angle, area, arc length, and perimeter.
3. Read radius r in meters and angle θ in degrees.
4. Convert radius: $R = r \times 100$.
5. Calculate area: $A = (\theta / 360) \times \pi \times R \times R$.
6. Calculate arc length: $L = (\theta / 360) \times 2 \times \pi \times R$.
7. Calculate perimeter: $P = L + 2 \times R$.

8. Display area, arc length, and perimeter.

9. Stop.

Algorithm evaluation: The radius is first converted from meters to centimeters so that all final values are in the required units. The standard sector formulas are then applied.

C Program

```
#include <stdio.h>
int main()    {
float r_m, r_cm, angle;
float area, arc_length, perimeter;
const float PI = 3.14159265;
printf("Enter radius in meters: ");
scanf("%f", &r_m); printf("Enter angle in degrees: ");
scanf("%f", &angle);
r_cm = r_m * 100;
area = (angle / 360.0) * PI * r_cm * r_cm;
arc_length = (angle / 360.0) * 2 * PI * r_cm;
perimeter = arc_length + 2 * r_cm;
printf("Area of sector = %.2f cm^2\n", area);
printf("Arc length = %.2f cm\n", arc_length);
printf("Perimeter of sector = %.2f cm\n", perimeter);
return 0;
}
```

Test Case 1: Input: 1 m, 90 deg

```
kali@kali: ~/hdoc/q2
File Actions Edit View Help
kali@kali:~/hdoc/q2$ gcc sector.c
kali@kali:~/hdoc/q2$ ./a.out
Enter radius in meters: 1
Enter angle in degrees: 90
Area of sector = 7853.98 cm^2
Arc length = 157.08 cm
Perimeter of sector = 357.08 cm
kali@kali:~/hdoc/q2$
```

Test Case 2: Input: 2 m, 180 deg

```
kali@kali: ~/hdoc/q2
File Actions Edit View Help
kali@kali:~/hdoc/q2$ gcc sector.c
kali@kali:~/hdoc/q2$ ./a.out
Enter radius in meters: 2
Enter angle in degrees: 180
Area of sector = 62831.85 cm^2
Arc length = 628.32 cm
Perimeter of sector = 1028.32 cm
kali@kali:~/hdoc/q2$
```

Test Case 3: Input: 0.5 m, 60 deg

```
kali@kali: ~/hdoc/q2
File Actions Edit View Help
kali@kali:~/hdoc/q2$ gcc sector.c
kali@kali:~/hdoc/q2$ ./a.out
Enter radius in meters: 0.5
Enter angle in degrees: 60
Area of sector = 1308.00 cm^2
Arc length = 52.36 cm
Perimeter of sector = 152.36 cm
kali@kali:~/hdoc/q2$
```

Conclusion

The C program was successfully compiled and executed in the Kali Linux environment. It correctly calculated the **area**, **arc length**, and **perimeter** of the sector after converting the radius from meters to centimeters. The actual outputs matched the expected outputs for all the prepared test cases. Therefore, the program is correct and all the test cases passed successfully.
